Sound quality in cochlear-implant users: measurement, signal processing and a deployed platform

Main PhD project • SOUND Lab, University of Cambridge • 2022–2026

Linked: JASA (2026) • Forum Acusticum (2025) • PhD thesis (not peer-reviewed)

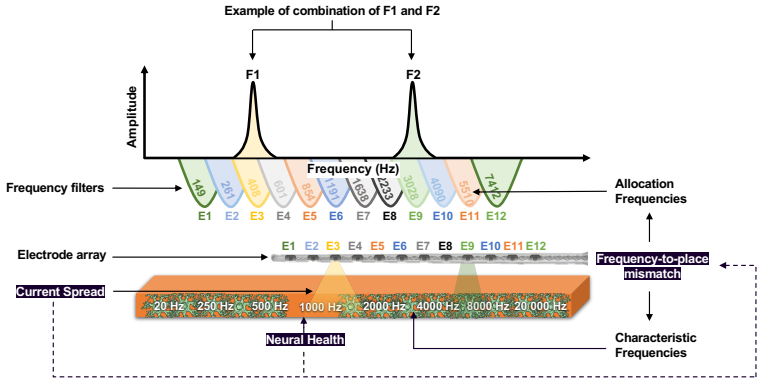
Which implant-related factors can still be tuned after switch-on to improve sound quality - and how do we measure their effect?

Approach

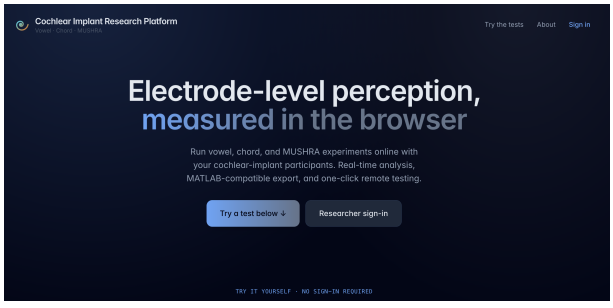
Cochlear-implant outcomes vary widely between users (speech-in-noise, sound quality, quality of life), partly because of individual, implant-related factors. This project targets the subset that can be manipulated after switch-on - frequency-to-place factors, *i.e.* how well spectral information is distributed and conveyed to the auditory system. The challenge was to design quick, ecological tasks that still probe effects at the level of individual electrodes; across many pilots I compared paradigms (paired comparisons, Likert ratings, an adaptive evolutionary procedure), post-processing (Bayesian ranking and machine learning), and signal types (speech and music, CI simulations, simplified vowels and chords).

Outcome

A clinically usable testing method, a browser-based platform that lets clinicians and researchers run the test with real-time analysis and remote testing, and working expertise in collecting and analysing perceptual data efficiently under tight participant constraints.



Experimental paradigm and the frequency-to-place factors investigated.



The online testing platform built for the project.